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REVIEW

Potassium-transporting proteins in skeletal muscle: cellular location and fibre-type differences

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Abstract
Potassium (K^+) displacement in skeletal muscle may be an important factor in the development of muscle fatigue during intense exercise. It has been shown *in vitro* that an increase in the extracellular K^+ concentration ($[K^+]_e$) to values higher than approx. 10 mM significantly reduce force development in unfatigued skeletal muscle. Several *in vivo* studies have shown that $[K^+]_e$ increases progressively with increasing work intensity, reaching values higher than 10 mM. This increase in $[K^+]_e$ is expected to be even higher in the transverse (T)-tubules than the concentration reached in the interstitium. Besides the voltage-sensitive K^+ (K_v) channels that generate the action potential (AP) it is suggested that the big-conductance Ca^{2+}-dependent K^+ ($K_{Ca}1.1$) channel contributes significantly to the K^+ release into the T-tubules. Also the ATP-dependent K^+ (K_{ATP}) channel participates, but is suggested primarily to participate in K^+ release to the interstitium. Because there is restricted diffusion of K^+ to the interstitium, K^+ released to the T-tubules during AP propagation will be removed primarily by reuptake mediated by transport proteins located in the T-tubule membrane. The most important protein that mediates K^+ reuptake in the T-tubules is the Na^+,K^+-ATPase α_2 dimers, but a significant contribution of the strong inward rectifier K^+ (Kir2.1) channel is also suggested. The Na^+, K^+, $2Cl^-$ 1 (NKCC1) cotransporter also participates in K^+ reuptake but probably mainly from the interstitium. The relative content of the different K^+-transporting proteins differs in oxidative and glycolytic muscles, and might explain the different $[K^+]_e$ tolerance observed.
Keywords channels, cotransporters, distribution, exercise, Na^+,K^+-ATPase.

Potassium (K^+) has long been proposed as an important determinant of muscle fatigue (Fenn 1940). At rest, the extracellular potassium concentration ($[K^+]_e$) is closely regulated between 3.8 and 5 mM (Sjøgaard *et al.* 1985, Choi *et al.* 2001), while the intracellular potassium concentration ($[K^+]_i$) is approx. 165 mM (Sjøgaard *et al.* 1985). It is impossible to maintain this K^+ gradient during strenuous physical activity. Studies have shown that $[K^+]_i$ can be reduced to values below 130 mM (Sjøgaard *et al.* 1985, Balog & Fitts 1996), and, depending on the experimental model used, $[K^+]_e$ has been shown to reach values of between 6.5 and 8.5 mM in venous blood draining from the working muscle (Vyskocil *et al.* 1983, Sjøgaard *et al.* 1985, Medbø & Sejersted 1990, Fallentien *et al.* 1992, Vøllestad *et al.* 1994, Bangsbo *et al.* 1996), and of >10 mM in the interstitium (Green *et al.* 1999, Juel *et al.* 2000b, Nordsborg *et al.* 2003). Lastly, Clausen (2008) calculated that $[K^+]_e$ could reach values between 26 and 52 mM depending on the fibre type used. *In vitro* studies have shown that $[K^+]_e$ levels above 10–12 mM lead to approx. 20 mV depolarization of the muscle fibre, followed by a corresponding reduction in force development (Juel 1986, Cairns *et al.* 1995, 1997, Nielsen *et al.* 2001, Pedersen *et al.* 2005). The underlying mechanism is thought to be a depolarization-induced inactivation of the voltage-sensitive sodium ($Na_v1.4$) channels present in the muscle fibre, where a depolarization as little as 10–15 mV has been shown to inactivate the $Na_v1.4$ channels (Hille 2001). This will significantly reduce the calcium (Ca^{2+}) release from the sarcoplasmic reticulum and ultimately cause muscle fatigue.

The change in K^+ homeostasis may be particularly pronounced in the transverse (T)-tubules because of their long and narrow shape (Franzini-Armstrong *et al.* 1975, Dulhunty 1984). Several groups have suggested or shown that the diffusion of K^+ to the interstitium is very small or nonexistent (Wallinga *et al.* 1999, Nielsen & Clausen 2000, Balog & Fitts 2001, Nielsen *et al.* 2004, Swift *et al.* 2006). Consequently, K^+ released to the T-tubule lumen can be removed only by reuptake mediated by transport proteins located in the T-tubule membrane. Taking into account the apparently lower relative content of Na^+,K^+-ATPase subunits in the T-tubule membrane compared with the cell surface membrane (Narahara *et al.* 1979, Venosa & Horowicz 1981, Williams *et al.* 2001, Nielsen *et al.* 2003a) and a relatively small T-tubule lumen, it seems reasonable to suggest that $[K^+]_e$ becomes higher in the T-tubules than in the interstitium. This is supported by the studies of Garcia *et al.* (1991) and Edman & Lou (1992), who showed that in extremely fatigued muscle fibres, only the peripheral myofibrils could contract during stimulation, while the myofibrils deeper in the muscle fibre were inactive. They suggested that this might be caused by an inhibition of action potential (AP) propagation along the T-tubule membrane because of an imbalance in the ion composition. In particular, an increase in $[K^+]_e$ could inhibit the excitation–contraction (E–C) coupling in the muscle fibre and ultimately cause muscle fatigue. In contrast, Cairns *et al.* (2009), found that in their protocol the surface rather than the T-tubular membrane is the site of impairment.

The transport of K^+ is mediated by a complex interplay of several K^+-transporting proteins located in the muscle fibre cell membrane. Therefore, to understand what causes the K^+ imbalance in the interstitium and the T-tubules during muscle fatigue, it is necessary to investigate which K^+-transporting proteins are present in skeletal muscle. Second, it is important to know both the absolute number of the different proteins present, their relative cellular distribution (T-tubule membrane vs. cell surface membrane), and the activity level of the K^+-transporting proteins present both at rest and during muscle contraction.

This review gives a broad overview of: (1) K^+-transporting proteins present in the cell surface and T-tubule membrane in skeletal muscle fibres; (2) the proposed relative cellular distribution of these proteins in the cell surface and T-tubule membrane; (3) differences in their distribution between different fibre types (oxidative/glycolytic); and (4) their proposed function in skeletal muscle.

K^+-transporting proteins in skeletal muscle

Comparing effect or localization of a protein may give rise to methodological problems. One problem is to compare *in vitro* and *in vivo* experiments, as reviewed by Bonen *et al.* (1994) or to compare between different species (see later). Another problem in physiological experiments could be the use of different statistical models, especially those involving repeated measures. This problem has been discussed earlier (Kristensen & Hansen 2004). Lastly, most measurements of protein content are not quantitative and give inadequate information about the recovery of the protein. It has been suggested that this could influence whether or not the data are representative (McKenna *et al.* 2003).

Several different proteins have been either shown or suggested to be involved in K^+ release and K^+ reuptake in skeletal muscle (for reviews, see Sejersted & Sjøgaard 2000, Clausen 2003, Jurkat-Rott *et al.* 2006, Allen *et al.* 2008). There are three main classes of K^+-transporting proteins in skeletal muscles. These are the channels, the cotransporters and the Na^+,K^+-ATPase. The structures of the different subunits making up the functional proteins are important both for the regulation of their activity and for classifying the proteins.

K^+ channels: protein structure and cellular location

Because the equilibrium potential for K^+ (E_K) in skeletal muscle is normally more negative than the resting membrane potential (E_m), the driving force for K^+ is outward and K^+ release is passively mediated by K^+ channels. The overall current (I) can be written as:

$$I = NP_o\gamma(E_m - E_K)$$

where N is the absolute number of channels, P_o is the probability of channel opening and γ is the single-channel current. Approximately 80 different K^+ channels have been identified in mammals (Hille 2001) but only few have been observed in skeletal muscle cell surface and T-tubule membrane (see below). These channels can be divided into three superfamilies depending on the number of transmembrane domains (TMDs): the Ca^{2+} (K_{Ca})- and voltage (K_v)-regulated K^+ channels (six TMDs), the constitutive K^+ channels [K_{2P} channels (four TMDs)] and the inward rectifier K^+ channels [Kir channels (two TMDs)].

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Six-TMD superfamily: K_v and K_{Ca} channels

At least two groups of channels from the six-TMD superfamily are present in skeletal muscle, the voltage-dependent K^+ (K_v) channels and the Ca^{2+}-dependent K^+ (K_{Ca}) channels.

K_v channels

K_v channels include traditional delayed rectifier channels, A-type (K_A) channels and channels that fall in between. This diversity is because of differences in the structural and functional classification (Hille 2001). The functional K_v channel is made of four six-TMD α subunits. The channel P_o is regulated via a voltage-sensitive gating system at TMD4 of the channel, also named S4 (Hille 2001). Furthermore, several auxiliary β subunits (MiRP1, MiRP2 and minK) have been shown to change the conductivity, gating kinetics and pharmacology of the functional channel (for reviews, see McCrossan & Abbott 2004, Li *et al.* 2006). Controversy continues over which channels from the K_v family are present in skeletal muscle. To date, 12 subfamilies of the K_v family (K_v1.x–K_v12.x) have been detected (for reviews, see Gutman *et al.* 2005, Li *et al.* 2006). Several of these subfamilies have been reported to be expressed in skeletal muscle on RNA level (for reviews, see Coetzee *et al.* 1999, Gutman *et al.* 2005, Jurkat-Rott *et al.* 2006). However, only K_v1.4, 3.4 and 7.3 seem to be functionally important in skeletal muscle cell membrane (for a review, see Jurkat-Rott *et al.* 2006), but their cellular distribution is unknown.

K_{Ca} channels

Ca^{2+}-dependent K^+ channels can be divided into five different subfamilies: K_{Ca}1.x–K_{Ca}5.x (for a review, see Wei *et al.* 2005). Sometimes these channels are also named according to their conductance capability: BK, SK and IK (big, small and intermediate conductance). The major Ca^{2+}-dependent potassium channel found in the skeletal muscle is the big-conductance Ca^{2+}-dependent K^+ (K_{Ca}1.1) channel (Fink *et al.* 1983, Lüttgau & Wettwer 1983, Jacquemond & Allard 1998, Mallouk & Allard 2000, Tricarico *et al.* 2005). The nomenclature of this channel is historically diverse, meaning that other names such as mSlo, $BKCa^{2+}$ and MaxiK refer to the same channel. The K_{Ca}1.1 channel is normally composed of two different subunits. An assembly of four α subunits forms the functional channel pore, while the β subunits work as modulators of the channel P_o (for a review, see Toro *et al.* 1998). However, experiments indicate that the β subunit might not be present in skeletal muscle (Tseng-Crank *et al.* 1996, Chang *et al.* 1997, Nimigean & Magleby 1999). The K_{Ca}1.1 channel is a member of the six-TMD superfamily, even though the α subunit has an extra TMD (TMD0; Meera *et al.* 1997). The P_o is increased in response to increased $[Ca^{2+}]_i$ and depolarization. However, even though the structure of K_{Ca}1.1 is closely related to that of the K_v channels and it behaves like these in response to depolarization, the K_{Ca}1.1 channel is not expected to have the same function as the K_v channels.

The K_{Ca}1.1 channel has been shown to be present in both the cell surface membrane (Jacquemond & Allard 1998, Mallouk & Allard 2000) and the T-tubule membrane (Knaus *et al.* 1995, Nielsen *et al.* 2003a, Tricarico *et al.* 2005). The K_{Ca}1.1 channel is relatively more abundant in the T-tubule than in the cell surface membrane (Nielsen *et al.* 2003a) (Table 1, Fig. 1), indicating that this channel may contribute significantly to the regulation of K^+ in the T-tubules, as was also suggested by Tricarico *et al.* (2005). This effect is especially pronounced in glycolytic muscles probably due to a splicing α isoform more sensitive to Ca^{2+} compared with the isoform present in oxidative muscles (Tricarico *et al.* 2005). A more exact localization for the K_{Ca}1.1 channel may be in clusters around the

Table 1 Suggested cellular distribution and fibre-type differences for different K^+ channels. (a) Cellular distribution of K^+ channels in mixed muscles. The relative content of protein in the cell surface membrane of mixed muscle is set to 100 (arbitrary units). The relative content of the relevant protein in the T-tubules is only indicated as either being higher (>100) or lower (<100) than the relative content in the cell surface membrane. ≫ indicates much higher. The distribution is based on Nielsen *et al.* (2003a) and Kristensen *et al.* (2006).
(b) Fibre-type distribution of K^+ channels in oxidative and glycolytic muscles. The distribution of K_{Ca}1.1 is based on a qualified guess (see text), K_{ATP} on Figure 2 and Kir2.1 on Kristensen *et al.* (2006) respectively. The content of protein in oxidative muscles was set to 100 (arbitrary scale) and the content in glycolytic muscles was calculated relative to this

	(a) Cellular distribution		(b) Fibre type distribution	
K^+ channel	Cell surface membrane	T-tubule membrane	Oxidative muscle	Glycolytic muscle
K_v	100	†	ND	ND
K_{Ca}1.1	100	>100	100	>100%
K_{ATP}	100	<100	100	~300%*
Kir2.1	100	>>100	100	~790%*
K_{2P}	ND	ND	ND	ND

ND, presence not determined.
†Present in skeletal muscle T-tubule, but relative distribution not determined.
*Significantly different from oxidative muscle.

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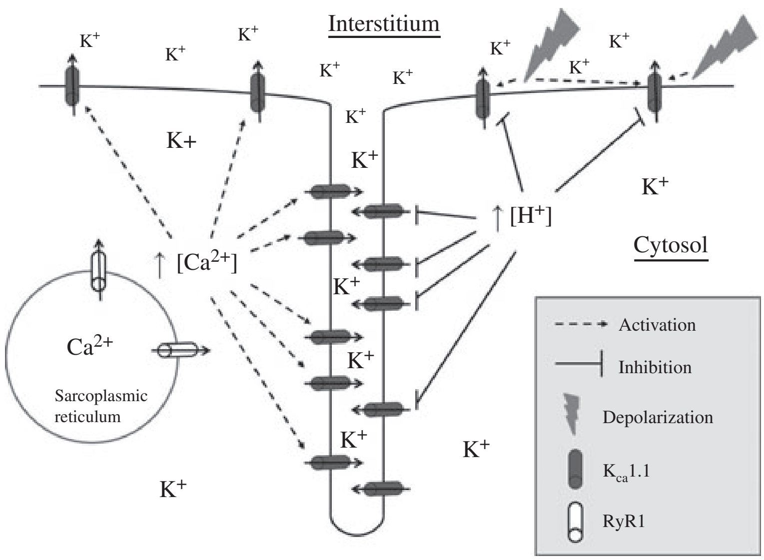

Figure 1 $K_{Ca}1.1$ channel cellular distribution and regulation. An overview of the cellular distribution and the different stimuli that either activate (increase P_o) or inhibit (decrease P_o) the $K_{Ca}1.1$ channels. RyR1, ryanodine receptor. For details and references, see text.

dihydropyridine (DHP) receptors, much like that shown for a similar Ca^{2+} channel in smooth muscles (Krummen *et al.* 2002). Because glycolytic fibres have three to five times more DHP receptors than oxidative fibres (Lamb & Walsh 1987, Delbono & Meissner 1996, Kristensen *et al.* 2006), such location would indicate that more $K_{Ca}1.1$ channels are present in glycolytic muscles than in oxidative muscles (Table 1). Even if the $K_{Ca}1.1$ channels are not clustered around the DHP receptors but are evenly dispersed in the T-tubule membrane, the higher number of T-tubules in glycolytic fibres compared with oxidative fibres (Cullen *et al.* 1984) still makes it probable that there is a higher relative content of $K_{Ca}1.1$ channels in glycolytic muscles.

The small-conductance Ca^{2+}-dependent K^+ ($K_{Ca}2.3$) channels are present in the cell surface membrane of denervated muscles, but not in normal innervated muscle (Roncarati *et al.* 2001, Jacobson *et al.* 2002). However, as little research has been carried out on these channels in skeletal muscle, the possibility that some of these channels are present in the cell surface or T-tubule membrane of normal skeletal muscle cannot be excluded.

Two-TMD superfamily: inward rectifier channels

Seven different subfamilies of the inward rectifier K^+ (Kir1.x–Kir7.x) channels have been observed (Kubo *et al.* 2005). Of these, at least the ATP-dependent K^+ (K_{ATP}) channel (Spruce *et al.* 1985, Davies 1990, Nielsen *et al.* 2003a) and the inward rectifier K^+ (Kir2.1) channel (Barrett-Jolley *et al.* 1999, Kristensen *et al.* 2006) are present in skeletal muscle.

K_{ATP} channel

The K_{ATP} channel is a weak inward rectifier channel. It is made up of four Kir6.x (kir6.1 or Kir6.2) subunits forming the functional channel pore and four regulatory sulfonylurea receptor (SUR) subunits (Babenko *et al.* 1998). This regulatory SUR subunit is not present in any of the other Kir channels (for reviews, see Coetzee *et al.* 1999, Seino 1999). Kir6.1 is expressed in skeletal muscle mitochondria (Suzuki *et al.* 1997), while the Kir6.2 is expressed in skeletal muscle cell surface and T-tubule membrane (Spruce *et al.* 1985, Nielsen *et al.* 2003a). The P_o is mainly regulated by the ATP:MgADP ratio (see later) and hence received the name K_{ATP} channel (Noma 1983).

The K_{ATP} channel is probably the most common K^+ channel in the cell surface membrane (Spruce *et al.* 1985). This means that even though the relative content of the K_{ATP} channel is higher in the cell surface membrane than in the T-tubule membrane (Nielsen *et al.* 2003a) (Table 1), it is possible that the absolute number of K_{ATP} channels in the T-tubule membrane is higher than the absolute number of $K_{Ca}1.1$ channels. This also means that the K_{ATP} channel can contribute considerably to K^+ regulation in the T-tubules. The relative content of the K_{ATP} channels in the cell surface membrane of mixed glycolytic muscles is approx. 2.4 times higher than in mixed oxidative muscles (Fig. 2; for methods and antibodies, see Nielsen *et al.* 2003a; Kristensen *et al.* 2006) (Table 1). The presence of a higher relative content of K_{ATP} channels in glycolytic muscles is supported by Matar *et al.* (2000), who observed a greater effect on extensor digitorum longus (EDL) muscles compared with soleus muscles when manipulating the K_{ATP} channel P_o. It may also explain the higher amount of interstitial K^+

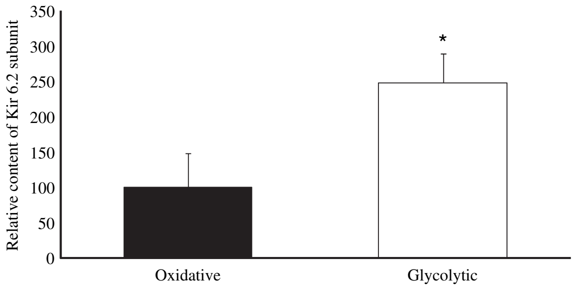

Figure 2 Distribution of the K_{ATP} channel Kir6.2 subunit in sarcolemma giant vesicles made from oxidative and glycolytic rat hind limb skeletal muscle. The content of protein in oxidative muscles was set to 100 (arbitrary scale) and the content in glycolytic muscles was calculated relative to this. *The difference in distribution of Kir6.2 is significant ($P < 0.0001$). Values are means ± SD. Sarcolemma giant vesicles were prepared according to Juel (1995), and the fibre-type specificity was prepared according to Kristensen *et al.* (2006).

found in EDL muscles (52.4 mM) compared with soleus muscles (26.0 mM) (Clausen 2008).

Kir2.1 channel

The Kir2.1 channel is a strong inward rectifier channel made of four Kir2.1 subunits. The strong inward rectification of the channel is caused by a voltage-dependent block of the channel pore by intracellular Mg^{2+} and polyamines that allows K^+ influx much more easily than K^+ efflux (Matsuda *et al.* 1987, Lopatin *et al.* 1994). This has led to the suggestion that this type of channel participates in the reuptake of K^+ (Wallinga *et al.* 1999, Sejersted & Sjøgaard 2000, Hille 2001).

Because of the apparently very low relative content of Kir2.1 channels in the cell surface membrane, they are suggested to be mainly located in the T-tubule membrane (Kristensen *et al.* 2006) (Table 1). The Kir2.1 channel is also much more abundant in glycolytic fibres compared with oxidative fibres (Kristensen *et al.* 2006) (Table 1), which makes sense considering the greater number of T-tubules in the glycolytic fibres (Cullen *et al.* 1984). However, the differences in the relative content of Kir2.1 between glycolytic and oxidative muscles are more pronounced than can be explained by the differences in the number of T-tubules. This indicates that the relative content of Kir2.1 channels is significantly higher in glycolytic muscle T-tubules than in oxidative muscles (Kristensen *et al.* 2006).

Four-TMD superfamily: constitutive K^+ channels

There exist 15 subfamilies of constitutive K^+ (K_{2P}) channels (Goldstein *et al.* 2005). These channels are made up of two identical four-TMD subunits, making a two-pore domain channel (Coetzee *et al.* 1999). Alternative names are the 2-P-K^+ channel and 'leaky' K^+ channels. So far, at least two isoforms have been detected at the mRNA level in skeletal muscle (Lesage *et al.* 1997, Kim *et al.* 2000). Channels from the four-TMD superfamily might therefore be present in skeletal muscle, although they are yet to be detected at the protein level, and, obviously, their contribution to K^+ release in skeletal muscle is unknown.

In nervous tissue, the 'leaky' K^+ current is very important for regulating the resting membrane potential and thereby neuronal excitability (for a review, see Talley *et al.* 2003). The channel has also been detected in smooth muscle (Koh *et al.* 2001), where the function seems to be the same as in neurones. If these channels are present in skeletal muscle, their main function could be identical with other excitable tissues, i.e. to participate in the regulation of K^+ homeostasis and E_m in the resting muscle.

K^+ channels: functional aspects

K^+ release

The amount of K^+ release is highly correlated with the frequency of APs and therefore with the activation of the K_v channels that are responsible for a major part of K^+ release (Sejersted & Sjøgaard 2000). The function of the K_v channels is to repolarize the membrane potential in the later part of the AP (for a review, see Sejersted & Sjøgaard 2000). This is mediated by a dramatic increase in the channel P_o. However, stimulation experiments have shown that the intracellular K^+ release can be up to twice as high as can be explained by the amount released by the K_v channels (Balog & Fitts 1996), indicating that other K^+ channels participate significantly in K^+ release, not just in the resting muscle fibre but also the contracting one.

The $K_{Ca}1.1$ channel has been proposed to be involved in K^+-induced muscle fatigue (Jacquemond & Allard 1998, Mallouk & Allard 2000). This is mainly because of its activation factors, including depolarization of the E_m and increasing $[Ca^{2+}]_i$ (Jacquemond & Allard 1998, Mallouk & Allard 2000) or stretching the cell membrane (Mallouk & Allard 2000) (Fig. 1). All these stimuli change during muscle contraction favouring an increase in the overall channel P_o. Another piece of evidence suggesting that the channel can be recruited during muscle contraction is the very low P_o shown in resting *in vitro* skeletal muscles (Allard *et al.* 1996), indicating that the channel has the potential to significantly increase the P_o. One contradictory finding, however, is that a drop in pH reduces the P_o (Laurido *et al.* 1991). This H^+ inhibition of the channel P_o is not competitive with Ca^{2+} binding (Laurido *et al.* 1991) and it is possible that this effect disappears when $[Ca^+]_i$

increases, as observed in smooth muscle (Schubert *et al.* 2001). *In vitro* stimulation experiments that showed a faster reduction in peak tetanic force in skeletal muscle incubated with a $K_{Ca}1.1$ channel activator compared with control muscles (Kristensen *et al.* 2006) suggest that the channel has the potential to contribute to K^+-induced muscle fatigue. However, isolated muscles studied *in vitro* have no blood flow, and it is therefore possible that the faster reduction in peak tetanic force during repetitive muscle contractions is due to an unphysiological increase in the $[K^+]_e$ around central fibres. Conversely, the fact that there is a restricted diffusion of K^+ from the T-tubule lumen to the interstitium and that the channel was shown to be relatively more abundant in the T-tubule than in the cell surface membrane (Nielsen *et al.* 2003a) (Table 1), reducing the probability that the result is caused by such an unphysiological increase in $[K^+]_e$. Consequently, the stimulation experiments, the cellular localization and the different stimuli that increase the P_o indicate that the channel participates significantly in K^+ release in skeletal muscle T-tubules during muscle contraction. It could be speculated that such participation in increasing $[K^+]_e$ might be a secondary effect of maintaining T-tubule membrane excitability. If there is a colocalization of the DHP receptor and the $K_{Ca}1.1$ channel, it is likely that the local $[Ca^{2+}]_i$ close to the $K_{Ca}1.1$ channel just below the membrane is higher than that measured in the cytosol. This could increase the channel P_o as observed in smooth muscles (Guia *et al.* 1999) and thereby assist repolarization of the T-tubule membrane during APs ensuring repriming of the Na^+ channels. Even without colocalization with the DHP receptor, the submembrane $[Ca^{2+}]_i$ may still increase owing to Ca^{2+} release from the sarcoplasmic reticulum: Jacquemond & Allard (1998) showed that the $K_{Ca}1.1$ channel P_o increased during muscle contraction without the presence of extracellular Ca^{2+}.

The K_{ATP} channel has also been implicated in fatigue when K^+ fluxes are involved (e.g. Davies *et al.* 1992, Light *et al.* 1994, Duty & Allen 1995, Gong *et al.* 2003, Cifelli *et al.* 2008). This association is mainly based on the different stimuli that change the channel P_o during repetitive muscle contractions (see below) combined with the very low *in vitro* P_o measured in resting muscle fibres (Spruce *et al.* 1985, Davies *et al.* 1992, McKillen *et al.* 1994, Allard *et al.* 1995). The K_{ATP} channel P_o increases in skeletal muscle when $[ADP]_i$ is increased in the presence of Mg^{2+} (Vivaudou *et al.* 1991, Allard & Lazdunski 1992), by increasing adenosine and GTP (Barrett-Jolley *et al.* 1996), and when $[ATP]_i$ (Vivaudou *et al.* 1991) and pH_i are decreased (Davies 1990, Davies *et al.* 1992, Standen *et al.* 1992, Allard *et al.* 1995) (Fig. 3). In addition, if the site of impairment during muscle contraction is at the surface rather than the T-tubular membrane, as suggested by Cairns *et al.* (2009), the high relative content of K_{ATP} channels in the cell surface membrane (Spruce *et al.* 1985) also supports the idea of this channel being an important regulator of K^+ homeostasis during muscle contractions. Lastly, the channel implication in fatigue has been supported by stimulation experiments where *in vitro* skeletal muscle incubated with the K_{ATP} channel activator pinacidil showed a reduced AP amplitude (Gong *et al.* 2003) and a faster reduction in peak tetanic force (Matar *et al.* 2000, Gong *et al.* 2003, Kristensen *et al.* 2006). The involvement of the K_{ATP} channel was supported using genetically modified Kir6.2 knockout mice in which pinacidil had no effect on the reduction in peak tetanic force (Gong *et al.* 2003). Reducing or completely inhibiting the K_{ATP} channel would be expected to slow down the reduction in peak tetanic

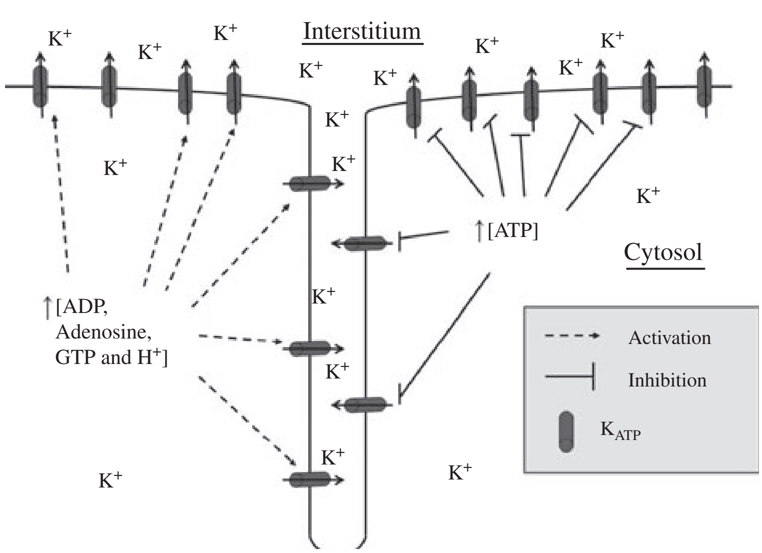

Figure 3 K_{ATP} channel cellular distribution and regulation. An overview of the cellular distribution and the different stimuli that either activate (increase P_o) or inhibit (decrease P_o) the K_{ATP} channels. For details and references, see text.

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force if the K_{ATP} channel is involved in K^+-induced muscle fatigue. However, this has not been observed when using different channel inhibitors or Kir6.2 knockout mice (Weselcouch *et al.* 1993, Light *et al.* 1994, Gong *et al.* 2000, 2003, Matar *et al.* 2000, Cifelli *et al.* 2007). It was suggested by Cifelli *et al.* (2007) that the lack of effect when inhibiting the K_{ATP} channel was due to a contractile dysfunction that was caused by increased $[Ca^{2+}]_i$. This was supported by Cifelli *et al.* (2008) who actually observed a slightly slower reduction in peak tetanic force in Kir6.2 mice when reducing Ca^{2+} influx. Another experiment that did not support the hypothesis of the K_{ATP} channel being an important regulator of K^+ homeostasis during muscle contraction was obtained by Nielsen *et al.* (2003a). Their *in vivo* study using microdialysis showed a significant reduction in potassium release from resting muscle when the K_{ATP} channel inhibitor glibenclamide was added to the perfusate, while no effect was observed when it was added during muscle contractions. This is in agreement with Lindinger *et al.* (2001), who concluded in an *in situ* study that most of the K^+ release from resting muscle takes place through the K_{ATP} channels, indicating a relatively high P_o for the K_{ATP} channel in the resting muscle. The apparent discrepancy observed in the K_{ATP} channel P_o might be related to the *in vitro* vs. *in vivo* approaches used in these studies. All the experiments showing a very low P_o were *in vitro* patch clamp studies (Spruce *et al.* 1985, Davies *et al.* 1992, McKillen *et al.* 1994, Allard *et al.* 1995), while the experiments suggesting a relatively high resting P_o were *in vivo* or *in situ* studies (Lindinger *et al.* 2001, 2002, Nielsen *et al.* 2003a). If the P_o for the K_{ATP} channel is lower in resting muscle *in vitro* (e.g. Matar *et al.* 2000, Kristensen *et al.* 2006) compared with *in vivo* (Lindinger *et al.* 2001, 2002, Nielsen *et al.* 2003a), the effect of adding an activator or inhibitor will be very different. Another problem with the *in vitro* stimulation experiments (Matar *et al.* 2000, Kristensen *et al.* 2006) is, as with the $K_{Ca}1.1$ channel, the lack of blood flow, meaning that the faster reduction in tetanic force development could be caused by an unphysiological increase in the $[K^+]_e$. This scenario is more likely to occur for the K_{ATP} channel than for the $K_{Ca}1.1$ channel, because the K_{ATP} channel is predominantly located in the cell surface membrane (Nielsen *et al.* 2003a) (Table 1) and because it is one of the most abundant K^+ channels in skeletal muscle (Spruce *et al.* 1985). Based on these experiments, direct involvement in muscle fatigue might not be the primary function of the K_{ATP} channel. Another suggested function for the channel during muscle contraction, and particularly at rest, is as a regulator of the energy status in skeletal muscle. It has been shown that the K_{ATP} channel has an influence on the basal as well as insulin-stimulated glucose transport in human skeletal muscle (Wasada *et al.* 2001), and a direct influence on glucose uptake in mouse skeletal muscle (Miki *et al.* 2002). It has also been suggested that the channel function both protects the muscle against ATP depletion during exercise and influences recovery of the resting muscle (Matar *et al.* 2000). Lastly, it has been shown by Gosmanov *et al.* (2004) that the K_{ATP} channel could be involved in hyperosmotic stimulation of the NKCC cotransporter (see later). These experiments support the idea that the channel is probably an important regulator of the E_m and energy status especially in resting muscles, although an important function in contracting muscles cannot be excluded.

K^+ reuptake

K^+ reuptake normally takes place against the driving force for K^+, and is therefore energy demanding, either as a secondary active transport via the Na^+-driven NKCC1 cotransporter, or primary active transport by the Na^+,K^+-ATPase (see later). For K^+ reuptake to occur through channels, the E_K has to become less negative than the E_m. An inward K^+ current independent of the Na^+,K^+-ATPase and the NKCC1 cotransporter has been measured in skeletal muscle (Hodgkin & Horowicz 1959, Standen & Stanfield 1978, Clausen & Overgaard 2000, Lindinger *et al.* 2001). This current is suggested to occur mainly across the T-tubule membrane, especially during muscle contraction (Wallinga *et al.* 1999). Under these conditions, E_K can potentially become less negative than E_m because of a large Cl^- conductance in the T-tubule membrane (Coonan & Lamb 1998). This Cl^- conductance acts to maintain a stable E_m during repetitive muscle contractions, while E_K will increase with increasing $[K^+]_e$, which could ultimately mean that the E_K at some point becomes more positive than E_m and that the driving force for K^+ turns inward, as suggested earlier (Coonan & Lamb 1998, Wallinga *et al.* 1999, Hille 2001, Allen *et al.* 2008). Wallinga *et al.* (1999) have set up a theoretical model for this function in skeletal muscle, and a similar mechanism has been shown in ventricular myocyte T-tubules (Clark *et al.* 2001).

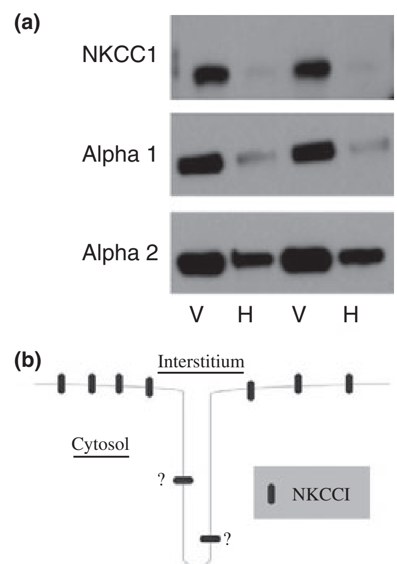
The Kir2.1 channel, which shows strong inward rectifying properties (Hagiwara *et al.* 1976, Eschke *et al.* 2002), could be partly responsible for this inward current. There are several indications supporting this suggestion. The first is the observation that the Kir2.1 channel is mainly located in the T-tubule membrane (Kristensen *et al.* 2006). Second, the channel P_o has been shown to be low at resting membrane potential in several types of tissue and has shown a voltage dependence resulting in an increased P_o in response to depolarization (Elam & Lansman 1995, Choe *et al.* 1999, Garneau *et al.* 2003). In addition, a decrease in extracellular pH combined with increased protein kinase A (PKA) activity has been shown to increase the channel P_o in non-muscular tissue (Malinowska *et al.* 2004). The functional role of the Kir2.1 channel could be as proposed by Wallinga *et al.* (1999): to withstand the reduction in skeletal muscle excitability of the T-tubules in the centre of the muscle fibre, promote E–C coupling and contribute to the prevention of muscle fatigue. This suggestion is indirectly supported by the experiments of Kristensen *et al.* (2006), who showed that Ba^{2+}, which reduces both the inward K^+ current (Standen & Stanfield 1978, Clausen & Overgaard 2000) and blocks the Kir2.1 channel (Owen *et al.* 1999) in a time- and dose-dependent manner, induced a faster reduction in tetanic force. This was suggested to be caused by a reduced K^+ uptake from the T-tubules, resulting in a faster increase in $[K^+]_e$. This mechanism could be particularly important in glycolytic muscles, because they contain more Kir2.1 channels than oxidative muscles (Table 1) and because the relative content of Na^+,K^+-ATPase in the T-tubule membrane is probably lower (see later).

Cotransporters

Protein structure and cellular location

Two types of cotransporters are involved in K^+ regulation: the $Na^+,K^+,2Cl^-$ (NKCC) and the $K^+,2Cl^-$ (KCC) cotransporters. KCC cotransporter mRNA has been reported to be expressed in skeletal muscle (Race *et al.* 1999), but it remains to be confirmed whether it has a functional role at the protein level. Two isoforms of the NKCC cotransporter are currently known: the NKCC1 cotransporter, which is present in almost all cell types including skeletal muscles, and the NKCC2 cotransporter, which is exclusively located in the kidney (for a review, see Russell 2000). The functional NKCC cotransporter is made up of one 12-TMD α subunit. TMD2 affects the cation affinity, while TMD4 and TMD7 affect the anion affinity (for a review, see Payne & Forbush 1995).

The NKCC1 cotransporter is present in the cell surface membrane of skeletal muscle (Kristensen *et al.* 2006). This corresponds with the findings of Wong *et al.* (1999), who used immunohistochemical staining to show the presence of the NKCC1 cotransporter at both the cell surface membrane and in T-tubule and/or intracellular membranes. The possible intracellular staining could represent pools of NKCC1 cotransporters available for translocation as observed for the NKCC2 in renal cells (Giménez & Forbush 2003). Although the study by Kristensen *et al.* (2006) suggested that the NKCC1 cotransporters were mainly located in the cell surface rather than in the T-tubule membranes, the exact relative distribution was not determined. However, when comparing the ratio of NKCC1 cotransporter present in cell surface membrane samples and homogenate samples with the corresponding ratios for the α_1 and α_2 Na^+,K^+-ATPase subunits (Fig. 4a), the pattern for the NKCC1 appears similar to that of α_1. Because the α_1 subunit is supposedly exclusively located in the cell surface membrane (see later), it is an obvious suggestion that this is also the case for the NKCC1 cotransporter in skeletal muscle (Fig. 4b). Based on the current studies we believe that the major part of the NKCC1 cotransporters is located in the cell surface membrane, but it cannot be excluded that a minor percentage is placed in the T-tubule



(c) Muscle type	NKCC1 Cotransporter
Oxidative	100
Glycolytic	˜180%*

Figure 4 Cellular and fibre-type distribution of the NKCC1 cotransporter in skeletal muscle. (a) Western blots used to calculate the relative cell surface membrane (*S*)/whole muscle (*W*) protein ratio (PR) using the following equation: $PR_X = ([S_X]/[W_X])$. Preparations of *S* were made as sarcolemma giant vesicles according to Juel (1991), while *W* were prepared from the remaining muscle tissue after the sarcolemma giant vesicle preparation. The tissues were homogenized, resuspended in Tris–SDS and subjected to Western blotting according to Kristensen *et al.* (2006). The protein amount was identical in all samples. X represents either the NKCC1 cotransporter, the α_1 or the α_2 Na^+,K^+-ATPase subunits. The relative *S*/*W* protein ratio was twice as high for the NKCC1 cotransporter as for the α_1 Na^+,K^+-ATPase subunit and six times higher than for the α_2 subunit ($n = 2$). (b) Suggested cellular distribution based on (a) and Kristensen *et al.* (2006). (c) Fibre-type distribution (Kristensen *et al.* 2006). *Significant differences.

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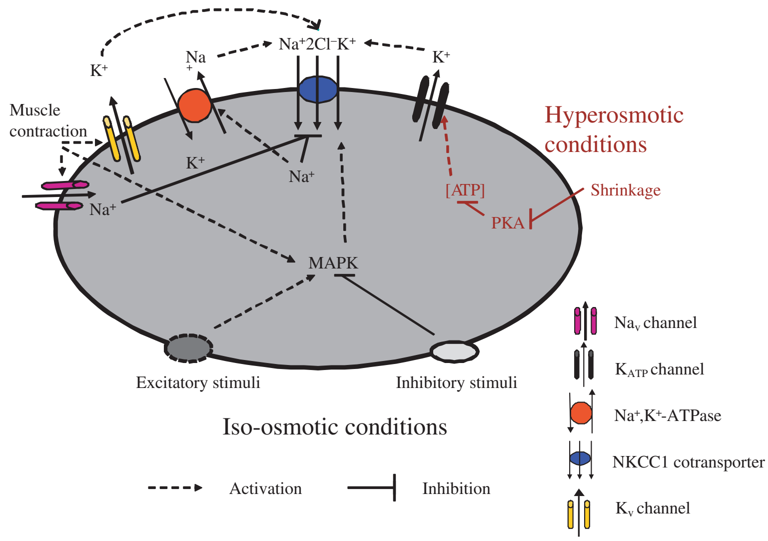

Figure 5 Regulatory mechanisms suggested for the NKCC1 cotransporter in skeletal muscle. Excitatory stimuli = adrenergic agonists. Inhibitory stimuli = insulin. For details and references, see text.

membrane. Furthermore, Kristensen *et al.* (2006) found a higher relative content of the NKCC1 cotransporter at the cell surface membrane of mixed glycolytic muscles than of mixed oxidative muscles (Fig. 4c) and Zhao *et al.* (2004) observed a higher relative content of NKCC1 cotransporters in EDL muscles than in soleus muscles.

Functional aspects

The NKCC cotransporter is electrically neutral and the transport activity therefore depends only on the concentration gradients for the transported ions. The main driving force for the cotransporter is the Na^+ gradient, which makes the NKCC cotransporter activity dependent on maintained Na^+,K^+-ATPase activity (Fig. 5) (Lindinger *et al.* 2002). Several different functions have been suggested for the NKCC1 cotransporter in skeletal muscle (for a review, see Gosmanov *et al.* 2003a). One possibility is regulation of K^+ homeostasis both at rest and during muscle contraction (Wong *et al.* 2001, Gosmanov *et al.* 2002a, Kristensen *et al.* 2006), and another is involvement in the regulatory volume increase (Lindinger *et al.* 2002, Gosmanov *et al.* 2003b, Zhao *et al.* 2004). Lastly, it has been suggested to influence $[Na^+]_i$ and thereby the activity of Na^+,K^+-ATPase both in adipocytes (Sargeant *et al.* 1995) and in skeletal muscle (Kristensen *et al.* 2006). The suggestions of the different functions of the NKCC1 cotransporter in skeletal muscle have often been based on activity measurements with $^{86}Rb^+$ transport as an indicator of a corresponding K^+ transport. However, a study by Dørup & Clausen (1994) indicated that this might not always be the case during bumetanide inhibition of the NKCC cotransporter. Furthermore, Dørup & Clausen (1996) found that there was no evidence for a bumetanide-sensitive K^+ transport in rat skeletal muscle. In contrast Lindinger *et al.* (2001) found that $^{42}K^+$ and $^{86}Rb^+$ fluxes correlated well in rat skeletal muscle both in time and magnitude using various NKCC cotransporter inhibitors. One explanation could be that the first studies (Dørup & Clausen 1994, 1996) were performed *in vitro*, while Lindinger *et al.* (2001) performed it *in situ*. There was also a difference in the bumetanide concentration. Dørup & Clausen (1994, 1996) mainly used a bumetanide concentration of 1 mM, while most other studies have been using concentrations between 10 and 100 μM (e.g. Wong *et al.* 1999, 2001, Lindinger *et al.* 2001, 2002, Gosmanov *et al.* 2002a,b, Zhao *et al.* 2004, Kristensen *et al.* 2006). To maintain a high specificity for the NKCC1 cotransporter a general rule is to use concentrations close to 10 μM (Russell 2000). It has been shown that high bumetanide concentrations can influence other anion transport systems such as the Cl^-/HCO_3^- exchanger (Gunn 1985, Jennings 1985) and Cl^- channels (Evans *et al.* 1986) in different tissues. However, it cannot explain the dose–response relationship for the effect of bumetanide on $^{42}K^+$ and $^{86}Rb^+$ observed in rat skeletal muscle (Dørup & Clausen 1994). It is assumed in the remaining of this review that $^{86}Rb^+$ is a reliable $^{42}K^+$ analogue for the NKCC1 cotransporter. However, it should be considered that this has been questioned by Dørup & Clausen (1994).

Kristensen *et al.* (2006) showed that inhibition of the NKCC1 cotransporter in soleus muscles induces an accelerated reduction in peak tetanic force during electrical stimulation, indicating that the NKCC1 cotransporter has an important function in ion regula-

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tion during muscle fatigue. This is supported by the observed increases in $^{86}Rb^+$ uptake (up to 33%) via the NKCC1 cotransporter measured during exercise, electrical stimulation and increased adrenaline concentration (Fig. 5, iso-osmotic conditions) (Wong *et al.* 2001, Gosmanov *et al.* 2002a). Further support of the proposition that the NKCC1 cotransporter is an important ion regulator during muscle activity is the 14% and 27% higher relative content of cotransporters observed in soleus and plantaris rat muscles, respectively, after training (Gosmanov *et al.* 2002a). These studies strongly indicate that the NKCC1 activity increases in the exercising muscle, and that this probably helps prevent the rise in $[K^+]_e$ and thereby postpones K^+-induced muscle fatigue. The signal cascade for the different NKCC stimuli (adrenaline, muscle contraction, etc.) under iso-osmotic conditions requires the extracellular signal-related kinase part of the mitogen-activated protein kinase pathway (Wong *et al.* 2001, Gosmanov & Thomason 2002, Gosmanov *et al.* 2002a,b). The possible roles of the NKCC1 cotransporter are summarized in Figure 5.

Another proposed function of the NKCC1 cotransporter in skeletal muscle is in volume regulation under hyperosmotic conditions (Fig. 5), as shown in both *in situ* and *in vitro* experiments (Lindinger *et al.* 2002, Gosmanov *et al.* 2003b, Zhao *et al.* 2004). Lindinger *et al.* (2002) suggested that the function of the NKCC1 cotransporter is to maintain cellular volume during exercise in non-contracting but hyperosmotically stimulated skeletal muscles, preventing them from shrinkage that has been shown to affect both contractility (Gulati & Babu 1982, Rapp *et al.* 1998) and cellular metabolism (Low *et al.* 1997). This increased NKCC1 activity could be caused, for example, by an increased adrenaline concentration or increased $[K^+]_e$ in the plasma during whole body exercise (Fig. 5). The K^+ activation of the cotransporter could also be autocrine induced by an increased P_o of the K_{ATP} channel because of a decrease in PKA activity (Fig. 5) (Gosmanov *et al.* 2004). This result could explain earlier findings showing that PKA activation suppresses hyperosmotic NKCC activation (Gosmanov *et al.* 2003b).

In conclusion, the NKCC1 cotransporter may have dual functional roles. In contracting muscle it participates in the reuptake of K^+ to reduce the increase in $[K^+]_e$ and thereby postpone muscle fatigue. However, in non-contracting muscles (during exercise), it regulates the muscle volume and prevents shrinkage of the muscle.

Na^+,K^+-ATPase

Na^+,K^+-ATPase is the main mediator of K^+ reuptake in skeletal muscle. It was discovered by Skou (1957), and the function of Na^+,K^+-ATPase in skeletal muscle has since been described in several reviews (Semb & Sejersted 1996, Nielsen & Clausen 2000, Sweeney & Klip 2001, Clausen 2003, Benziane & Chibalin 2008). The function of Na^+,K^+-ATPase in skeletal muscle is to maintain ion distribution and thereby muscle membrane potential and muscle excitability (for a review, see Clausen 2003). This is achieved by actively extruding three Na^+ ions from the muscle fibre and taking up two K^+ ions from the interstitium or T-tubule lumen. Na^+,K^+-ATPase consists of three different subunits, the α, β and FXYD subunits. The 10-TMD α subunit (110 kDa) is the catalytic subunit, comprising the binding sites for Na^+, K^+ and ATP (for a review, see Clausen 2003), and contains several phosphorylation sites (Al-Khalili *et al.* 2004). Four different α isoforms (α_{1-4}) have been described (for a review, see Blanco & Mercer 1998). The α_1 and α_2 isoforms have been thoroughly investigated, and have been conclusively shown to be present and functional in both human and rat skeletal muscle (e.g. Hundal *et al.* 1994, Juel *et al.* 2000a, 2001, Hansen 2001, Fowles *et al.* 2004). More discrepancy exists regarding the α_3 isoform in rat skeletal muscle. Some studies were unable to detect the α_3 on RNA level (Young & Lingrel 1987, Jewell *et al.* 1992, Gick *et al.* 1993), while others were able to do this (Murphy *et al.* 2006, 2008a). However, a functional α_3 isoform is still to be confirmed. In human skeletal muscle the α_3 isoform has been detected on both RNA (Murphy *et al.* 2004) and protein levels (Hundal *et al.* 1994, Murphy *et al.* 2004). Lastly, the α_4 subunit might be present, at least in human skeletal muscle (Keryanov & Gardner 2002). The β subunit is a 55 kDa one-TMD subunit thought to be required for proper folding and stabilization of Na^+,K^+-ATPase in the cell membrane, and may also influence the enzymatic properties of the α subunit (Hasler *et al.* 1998). Three different β isoforms (β_{1-3}) have been described (for a review, see Blanco & Mercer 1998), and all three seem to be present in rats (Arystarkhova & Sweadner 1997, Juel *et al.* 2000a, Fowles *et al.* 2004). Their presence in human skeletal muscle is more controversial. Hundal *et al.* (1994) and Juel *et al.* (2000a) were unable to detect the β_2 subunit at the protein level, whereas its existence at both protein and mRNA levels was reported by Murphy *et al.* (2004). The α and β subunits must be present in a 1 : 1 ratio as a dimer to form a functional Na^+,K^+-ATPase, while the pump can function without the FXYD subunit (Crambert *et al.* 2000). The FXYD subunit is a 15 kDa one-TMD subunit that appears to have the principal function of regulating the Na^+,K^+-ATPase affinity especially for Na^+, but also for K^+ (Crambert *et al.* 2002, Bibert *et al.* 2008). There are seven different FXYD isoforms, all of which are expressed in a tissue-specific manner (for reviews, see Sweadner & Real 2000, Garty & Karlish 2006), with

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FXYD1 (phospholemma, PLM) being the only one expressed in skeletal muscle. Further description of the cellular and fibre-type distributions of these components will include only the α_1, α_2, β_1, β_2 and PLM subunits and will be based mainly on studies in rats.

Alpha subunit

The cellular distribution of the Na^+,K^+-ATPase α_1 subunit has been determined in rat EDL and tibialis anterior muscles (Williams *et al.* 2001), in rat white gastrocnemius and a mixture of oxidative rat muscles (Lavoie *et al.* 1995, 1997), and in human soleus muscle (Hundal *et al.* 1994). These studies showed that the α_1 subunit is nearly exclusively located in the cell surface membrane (Table 2). Kristensen *et al.* (2008) showed that at least 11% and 17% of the α_1 subunits in EDL and soleus muscles, respectively, were colocalized with caveolin-3 (Cav-3) and therefore located in the membrane of caveolae (Table 2). There is a higher relative content of α_1 subunits in soleus than in EDL muscles (Fowles *et al.* 2004) and generally in oxidative muscles compared with glycolytic muscles (Thompson & McDonough 1996, Juel *et al.* 2001) (Table 3).

The distribution pattern for the α_2 subunit differs in several ways from the α_1 subunit. First, it has been shown in both rat and human skeletal muscle that the α_2 subunits are dispersed throughout the muscle fibre and are therefore located both in cell surface, T-tubule and perhaps intracellular membranes (Hundal *et al.* 1994, Lavoie *et al.* 1995, 1996, 1997, Williams *et al.* 2001).

Table 2 Suggested cellular distribution of the Na^+,K^+-ATPase α_1 and α_2 subunits in rat soleus and extensor digitorum longus (EDL) muscles, and in mixed muscles from sarcolemma giant vesicles (SGV). The cell surface membrane is divided into the sarcolemma membrane and caveolae membrane

	Sarcolemma membrane	Caveolae membrane	T-tubule membrane
α_1 (%)			
Soleus†	83	17	0
EDL†	89	11	0
SGV	86	14	*
α_2 (%)			
Soleus†	>33	17	<50
EDL†	42	5	53
SGV	84	16	*

*T-tubule membrane not present.
†Not all Cav-3 was immunoprecipitated from soleus and EDL muscles. It is therefore likely that some α subunits associated with the Cav-3 were not immunoprecipitated. This indicates that the relative content of α in the caveolae is probably 3–10% higher than suggested in both EDL and soleus muscles (Kristensen *et al.* 2008). For further references, see text.

Table 3 Relative distribution of α_1, α_2, β_1, β_2 and PLM subunits in soleus and EDL rat muscle homogenate (Western blotting). The content of the different proteins/subunits was set to 100 (arbitrary scale) in soleus muscles and the content in the EDL calculated relative to this. Representative data for α_1, α_2, β_1 and β_2 subunit distributions were obtained from Fowles *et al.* (2004), while the PLM values were from Rasmussen *et al.* (2008)

Protein	Soleus muscle	EDL muscle
α_1	100	~25%*
α_2	100	~120%
β_1	100	~40%*
β_2	100	~250%*
PLM	100	~95%

PLM, phospholemma; EDL, extensor digitorum longus.
*Significantly different from soleus muscles.

Second, although some studies have found a higher relative content of α_2 subunits in EDL than from soleus muscles (Clausen *et al.* 1982, Everts & Clausen 1992), other studies did not observe any significant difference in EDL and soleus muscles (Chin & Green 1993, Fowles *et al.* 2004) or in oxidative muscle compared with glycolytic muscle in general (Hundal *et al.* 1993, Thompson & McDonough 1996) (Table 3). The lack of differences in the relative content of α_2 subunits in homogenates from the two muscle types is unexpected because there are approx. 50% more T-tubules in EDL muscles than in soleus muscles (Cullen *et al.* 1984). This indicates that the relative content of the α_2 subunit in the soleus cell surface membrane is higher than in the corresponding EDL cell surface membrane. This fibre type-specific distribution pattern is confirmed by Fowles *et al.* (2004) and Juel (2009) who showed that the relative cell surface content of α_2 subunits were approx. 25% lower in EDL muscles compared with soleus muscles, and approx. 30% lower in mixed glycolytic muscles compared with mixed oxidative muscles. The cell surface/T-tubule ratio of the α_2 subunit in EDL muscles is approx. 1 (Nielsen *et al.* 2006). Of the α_2 subunits located in the cell surface membrane at least 5% are suggested to be located in the caveolae membrane (Kristensen *et al.* 2008). The relative content of α_2 subunits present in the cell surface membrane in soleus muscles has not been determined. However, based on the finding that the relative cell surface content of α_2 subunits is higher in soleus muscles compared with EDL muscles and assuming that the relative content of α_2 subunits is approximately the same in soleus and EDL muscles (Chin & Green 1993, Fowles *et al.* 2004), it is suggested that more than 50% of the α_2 subunits are located in the cell surface membrane in soleus muscles. Of these at least 17% are suggested to be located in the caveolae membrane (Kristensen *et al.* 2008). The suggested α_2 distribution is shown in Table 2.

Besides the differences in cellular distribution between the α_1 and α_2 subunits, there is also a difference in the relative content of the two subunits. It has been shown that there are approximately two to five times more α_2 subunits than α_1 subunits present in rat skeletal muscle (Sweadner *et al.* 1992, Lavoie *et al.* 1997, Hansen 2001) and approximately four to six times more α_2 subunits than α_1 subunits in human skeletal muscle (He *et al.* 2001, Golovina *et al.* 2003). The α_2/α_1 ratio probably increases with age (Sweadner *et al.* 1992, Al-Khalili *et al.* 2004). Due to the higher relative content of the α_2 subunit it is obvious that the Na^+,K^+-ATPase α_2 dimers contribute relatively more to the K^+ regulation compared with the α_1 dimers.

Beta subunit

Both β_1 and β_2 subunits have been shown to be present in rat skeletal muscle cell surface, T-tubule and perhaps intracellular membranes (Hundal *et al.* 1992, Lavoie *et al.* 1996, 1997). Lavoie *et al.* (1997) found that although the relative content of β_1 subunits was approximately four times higher than the relative content of β_2, the surface/(intracellular and probably T-tubule) membrane ratio for both β_1 and β_2 subunits in oxidative muscles was approx. 6.5, suggesting that the relative content of β subunits is much higher in the cell surface membrane than in the T-tubule membrane. They also found that the $\alpha:\beta$ ratio was 0.2 in the surface membrane and 0.4 in the intracellular and T-tubule membrane, indicating a large surplus of β subunits in both cell surface and T-tubule membranes. The cell surface membrane can be compartmentalized into sarcolemma and caveolae membranes. It has been shown in rat cardiac myocytes that more than 80% of the β_1 subunits were located in caveolae-containing membranes (Liu & Askari 2006). It is an obvious suggestion that a similar pool of the excess β subunits could be present in the caveolae membrane of skeletal muscle. The functional significance of this surplus of β subunits is unknown. The pattern is probably different in glycolytic muscles because there is a clear fibre-type difference for the β subunits. β_1 is present in a higher concentration in soleus muscles than in EDL muscles and in oxidative muscles compared with glycolytic muscles in general, while the β_2 subunit shows the opposite pattern (Hundal *et al.* 1993, Juel *et al.* 2001, Fowles *et al.* 2004, Zhang *et al.* 2006) (Table 3).

Phospholemma subunit

Immunofluorescence experiments have shown that the PLM subunit is located in both cell surface, T-tubule and perhaps intracellular membranes (Rasmussen *et al.* 2008). Furthermore, there is no significant difference in the relative content of the PLM subunit in homogenate from rat soleus muscles compared with EDL muscles (Rasmussen *et al.* 2008) (Table 3). However, Juel (2009) found that the relative content of PLM in the cell surface membrane was approximately twice as high in mixed oxidative muscles as in mixed glycolytic muscles.

Na^+,K^+-ATPase activity in skeletal muscle

It is not known whether the α and β subunits aggregate selectively (for instance, α_1 with β_1) or whether they are randomly aggregated. The cellular and fibre-type differences for α and β (and PLM) subunits are expected to have a significant influence on Na^+ and K^+ affinity and V_{max} of Na^+,K^+-ATPase. In *Xenopus* oocytes it has been shown that Na^+,K^+-ATPase dimers with α_1 have a high affinity for both Na^+ and K^+ compared with dimers with α_2 (Crambert *et al.* 2000, Bibert *et al.* 2008). Likewise, dimers with β_1 have a higher affinity for Na^+ than dimers with β_2 (Crambert *et al.* 2000).

The selective distribution of α_1 subunits in skeletal muscle implies that the T-tubules only possess $\alpha_2\beta_1$ and $\alpha_2\beta_2$ dimers. Because dimers with α_1 are expected to have a higher Na^+ affinity than dimers with α_2 and because the cell surface membrane contains all four combinations, it is suggested that there is a lower ion affinity in the T-tubules compared to the cell surface (Juel 2009). In soleus (representing oxidative muscle), the predominant combinations include β_1, and in EDL (representing glycolytic muscle), the predominant combinations include β_2 (see Table 3). Based on the cellular distribution of α subunits in soleus and EDL muscles, respectively (see Table 3), the Na^+ and K^+ affinity of Na^+,K^+-ATPase is expected to be higher in soleus than in EDL muscles. This has been confirmed in resting muscles, where the pump affinity for Na^+ is much higher in oxidative muscles than in glycolytic muscles (Juel 2009). However, the physiological implication of this difference is not clear and is further complicated by the finding that all the fibre type-dependent pump affinity differences disappear after exercise (Juel 2009). There could be several underlying mechanisms explaining this. One could be the contraction-induced translocation observed for both α and β subunits (Tsakiridis *et al.* 1996, Juel *et al.* 2000a, 2001, Sandiford *et al.* 2005), although this translocation could not be detected by McKenna *et al.* (2003) and Murphy *et al.* (2008b). Another mechanism could be an influence of the PLM subunit. This subunit has been shown to translocate to the cell surface membrane and to change its relative association with the Na^+,K^+-ATPase α subunit (Rasmussen *et al.* 2008). When non-phosphorylated PLM is associated with Na^+,K^+-ATPase in *Xenopus* oocytes, the pump affinity for both Na^+ and K^+ ions is decreased (Crambert *et al.* 2002, Bibert *et al.* 2008). However, when PLM is phosphorylated, Na^+,K^+-ATPase affinity for Na^+ is similar to that observed when PLM is not present but with a marked increase in V_{max} (Bibert *et al.* 2008). The degree of PLM phosphorylation in skeletal muscle is still under investigation. Rasmussen *et al.* (2008) did not find any changes in the degree of phosphorylation of PLM Ser68 in rat skeletal muscle. However, phosphorylation at other possible phosphorylation sites on PLM cannot be excluded. Exercise-induced phosphorylation on both Ser63 and Ser68 has been shown in human skeletal muscle (M. Thomassen, A.J. Rose, N.R. Andersen, L. Bune, D.J.A. Hanskov, J. Bangsbo and N. Nordsborg, unpublished results) which suggests that phosphorylation has a regulatory effect on Na^+,K^+-ATPase during muscle contraction.

Concluding remarks and perspectives

K^+ displacement in skeletal muscle may be an important factor in the development of muscle fatigue during intense exercise. Based on the relative cellular distribution (T-tubule membrane vs. cell surface membrane) of the K^+-transporting proteins, it is suggested that the Na^+,K^+-ATPase α_2 dimers, the $K_{Ca}1.1$ and Kir2.1 channels play a significant role in regulating K^+ fluxes across the T-tubule membrane, while the Na^+,K^+-ATPase α_1 and α_2 dimers, the K_{ATP} channel and the NKCC1 cotransporter are important in regulating K^+ fluxes across the cell surface membrane. Combining the cellular protein localization with different *in vitro* stimulation experiments we suggest that especially the Na^+,K^+-ATPase α_2 dimers, the $K_{Ca}1.1$ and Kir2.1 channels are important in K^+ regulation during muscle fatigue, while the Na^+,K^+-ATPase α_1 dimers and the NKCC1 cotransporter seem less important. Due to the contradictory results, it is not possible to decide how important the K_{ATP} channel is in the contribution to K^+-induced muscle fatigue.

The proposal for the involvement of the different K^+-transporting proteins is primarily based on experiments that compare relative protein amounts or use *in vitro* stimulation experiments. To measure the absolute amount of all the K^+-transporting proteins and to establish the *in vivo* protein activity both in surface and T-tubule membranes will be of considerable significance. Knowing these parameters makes it possible to calculate the exact amount of K^+ released or taken up by the different proteins and thereby determine the contribution from each K^+-transporting protein. However, a major problem is the accessibility of the T-tubules *in vitro*, but especially *in vivo*. One possible technique is to use *in vivo* cloning. By fusing a fluorescent protein (e.g. EGFP) to the protein of interest, it is possible to investigate both cellular localization and co-localization between different proteins. Another interesting method could be to use single fibres – either whole fibres or skinned fibres with an intact T-tubular system.

K^+-induced muscle fatigue is still far from being understood and there is a great deal of work to be done to elucidate all the facets of this phenomenon. The topic is further complicated because the K^+ regulation in muscle fibres is influenced by other mechanisms, such as Cl^- channel activity and Na^+ concentration changes. The use of old and newly developed techniques will increase our understanding of the different facets of K^+-induced muscle fatigue.

Conflict of interest

There is no conflict of interest.

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